

Code:

```
#include <stdio.h>

void calculateSpan(int price[], int n, int S[]) {
    int st[1005];
    int top = -1;

    // Span value of first day is always 1
    S[0] = 1;
    st[++top] = 0;

    // Calculate span values for remaining days
    for (int i = 1; i < n; i++) {
        while (top >= 0 && price[st[top]] <= price[i]) {
            top--;
        }

        S[i] = (top == -1) ? (i + 1) : (i - st[top]);
        st[++top] = i;
    }
}

void printArray(int arr[], int n) {
    for (int i = 0; i < n; i++) {
        printf("%d", arr[i]);
        if (i < n - 1) printf(" ");
    }
    printf("\n");
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int price[1005];
    for (int i = 0; i < n; i++) {
        scanf("%d", &price[i]);
    }

    int S[1005];
    calculateSpan(price, n, S);
}
```

```
    printArray(S, n);  
  
    return 0;  
}
```

Input:

```
6  
9 8 7 6 5 4
```

Output:

```
1 1 1 1 1 1
```